

Assignment R programming

2026-09-22

R Markdown

This is an R Markdown document. Markdown is a simple formatting syntax for authoring HTML, PDF, and MS Word documents. For more details on using R Markdown see <http://rmarkdown.rstudio.com>.

When you click the **Knit** button a document will be generated that includes both content as well as the output of any embedded R code chunks within the document. You can embed an R code chunk like this:

```
student_data <- read.csv("C:/Users/DELL/Downloads/Student-Data-New-20.csv")
student_data
```

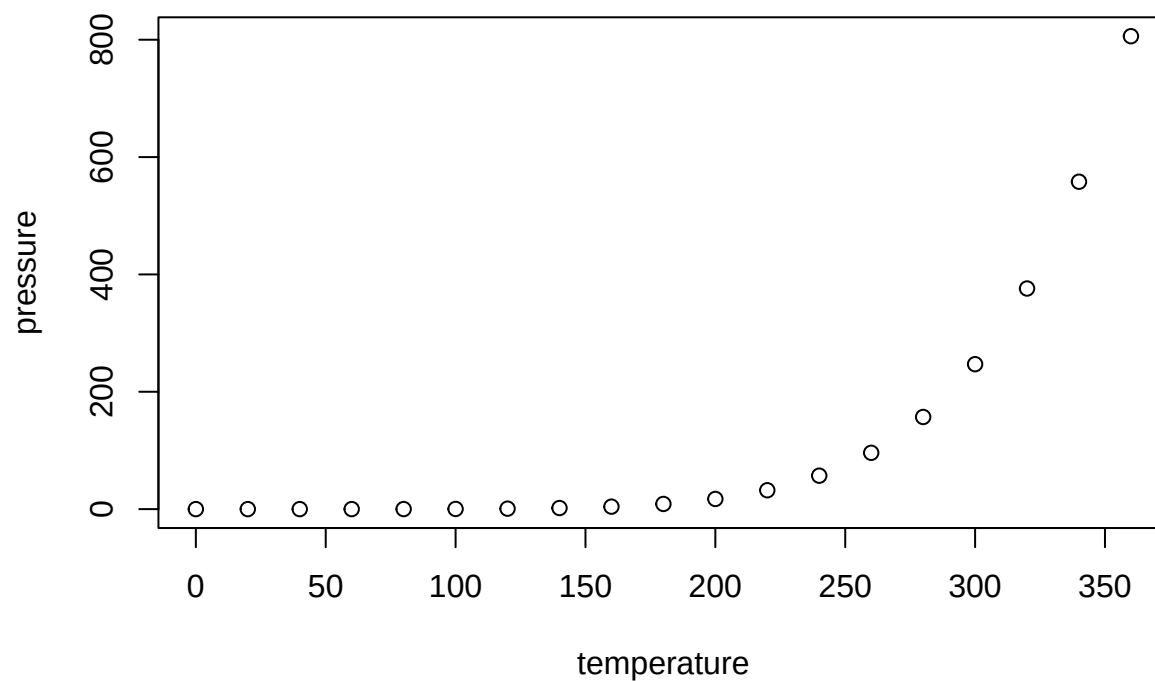
##	Student_ID	Name	Gender	Age	Marks	Attendance	Result
## 1	S202401	Mahika	Female	22	40	70	Pass
## 2	S202402	Saanvi	Female	22	47	89	Pass
## 3	S202403	Kiara	Female	20	64	81	Pass
## 4	S202404	Atharv	Male	21	75	58	Pass
## 5	S202405	Rohan	Male	22	84	67	Pass
## 6	S202406	Vihaan	Male	22	54	66	Pass
## 7	S202407	Sara	Female	20	82	94	Pass
## 8	S202408	Aaradhya	Female	22	93	77	Pass
## 9	S202409	Pari	Female	22	43	93	Pass
## 10	S202410	Anika	Female	20	86	86	Pass
## 11	S202411	Ishaan	Male	20	64	65	Pass
## 12	S202412	Myra	Female	19	86	32	Fail
## 13	S202413	Siya	Female	19	94	96	Pass
## 14	S202414	Mohammed	Male	22	69	67	Pass
## 15	S202415	Reyansh	Male	19	79	76	Pass
## 16	S202416	Ananya	Female	20	43	20	Fail
## 17	S202417	Navya	Female	19	92	73	Pass
## 18	S202418	Aditya	Male	19	54	58	Pass
## 19	S202419	Krishna	Male	19	97	70	Pass
## 20	S202420	Aryan	Male	21	45	75	Pass

```
head(student_data)
```

##	Student_ID	Name	Gender	Age	Marks	Attendance	Result
## 1	S202401	Mahika	Female	22	40	70	Pass
## 2	S202402	Saanvi	Female	22	47	89	Pass
## 3	S202403	Kiara	Female	20	64	81	Pass
## 4	S202404	Atharv	Male	21	75	58	Pass
## 5	S202405	Rohan	Male	22	84	67	Pass
## 6	S202406	Vihaan	Male	22	54	66	Pass

Including Plots

You can also embed plots, for example:



```
summary(student_data)
```

```
##      Student_ID      Name      Gender      Age      Marks
## Length   :20   Length   :20   Length   :20   Min.    :19.0   Min.    :40.00
## N.unique :20   N.unique :20   N.unique : 2   1st Qu.:19.0   1st Qu.:52.25
## N.blank  : 0   N.blank  : 0   N.blank  : 0   Median :20.0   Median :72.00
## Min.nchar: 7   Min.nchar: 4   Min.nchar: 4   Mean    :20.5   Mean    :69.55
## Max.nchar: 7   Max.nchar: 8   Max.nchar: 6   3rd Qu.:22.0   3rd Qu.:86.00
##                                     Max.    :22.0   Max.    :97.00
##      Attendance      Result
## Min.    :20.00   Length   :20
## 1st Qu.:65.75   N.unique : 2
## Median :71.50   N.blank  : 0
## Mean    :70.65   Min.nchar: 4
## 3rd Qu.:82.25   Max.nchar: 4
## Max.    :96.00
```

```
str(student_data)
```

```
## 'data.frame':   20 obs. of  7 variables:
## $ Student_ID: chr  "S202401" "S202402" "S202403" "S202404" ...
```

```
## $ Name      : chr  "Mahika" "Saanvi" "Kiara" "Atharv" ...
## $ Gender    : chr  "Female" "Female" "Female" "Male" ...
## $ Age       : int   22 22 20 21 22 22 20 22 22 20 ...
## $ Marks     : int   40 47 64 75 84 54 82 93 43 86 ...
## $ Attendance: int   70 89 81 58 67 66 94 77 93 86 ...
## $ Result    : chr  "Pass" "Pass" "Pass" "Pass" ...
```

```
colSums(is.na(student_data))
```

```
## Student_ID      Name      Gender      Age      Marks Attendance      Result
##           0           0           0           0           0           0           0
```

```
median(student_data$Marks)
```

```
## [1] 72
```

```
var(student_data$Marks)
```

```
## [1] 387.8395
```

```
sd(student_data$Marks)
```

```
## [1] 19.69364
```

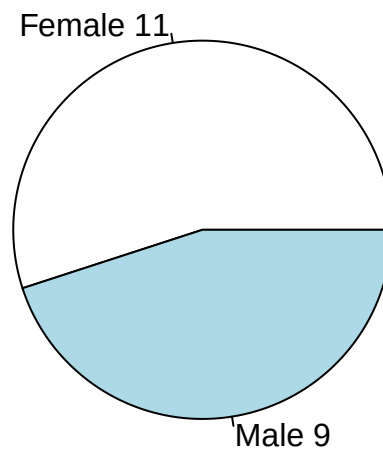
```
gender_count <- table(student_data$Gender)
```

```
gender_count
```

```
##
## Female      Male
##        11         9
```

```
pie(
  gender_count,
  main = "Gender Distribution",
  labels = paste(
    names(gender_count),
    gender_count
  )
)
```

Gender Distribution



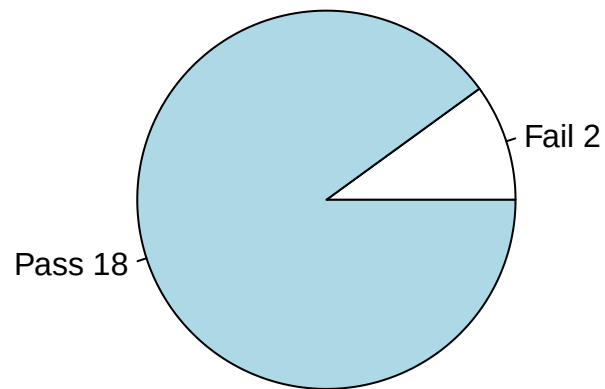
```
result_count <- table(student_data$Result)
```

```
result_count
```

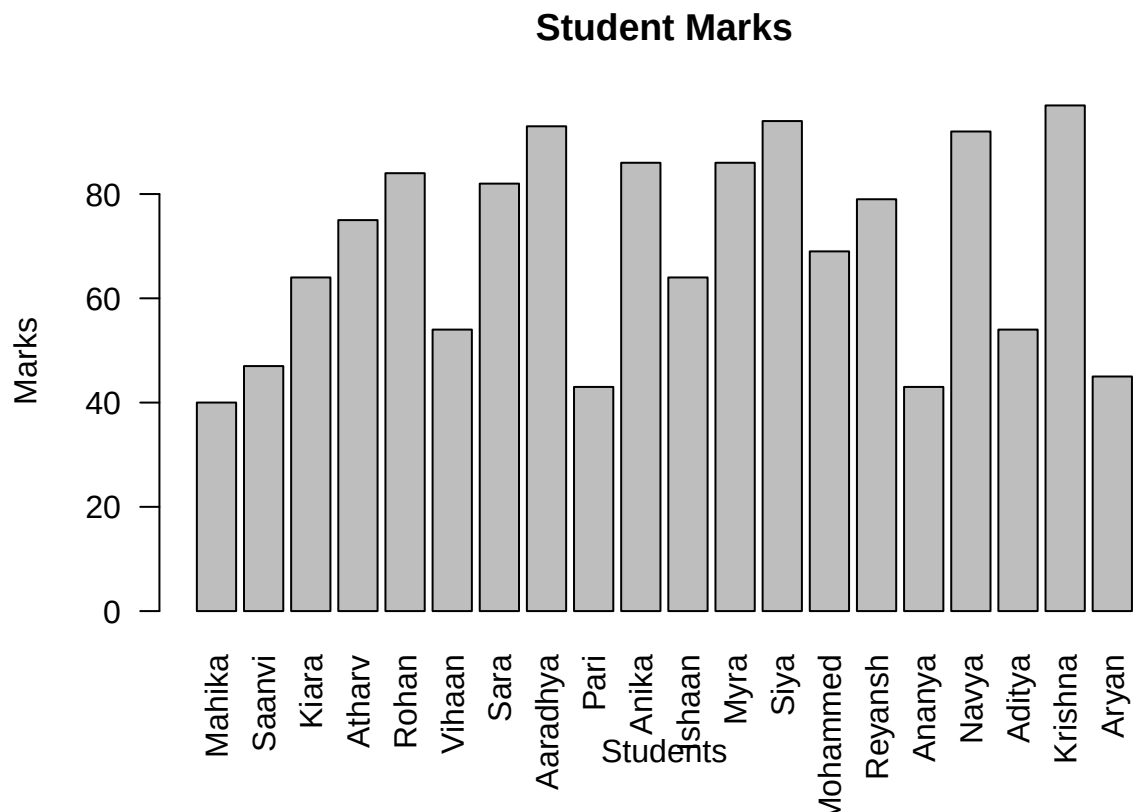
```
##  
## Fail Pass  
##    2   18
```

```
pie(  
  result_count,  
  main = "Student Result Distribution",  
  labels = paste(  
    names(result_count),  
    result_count  
  )  
)
```

Student Result Distribution



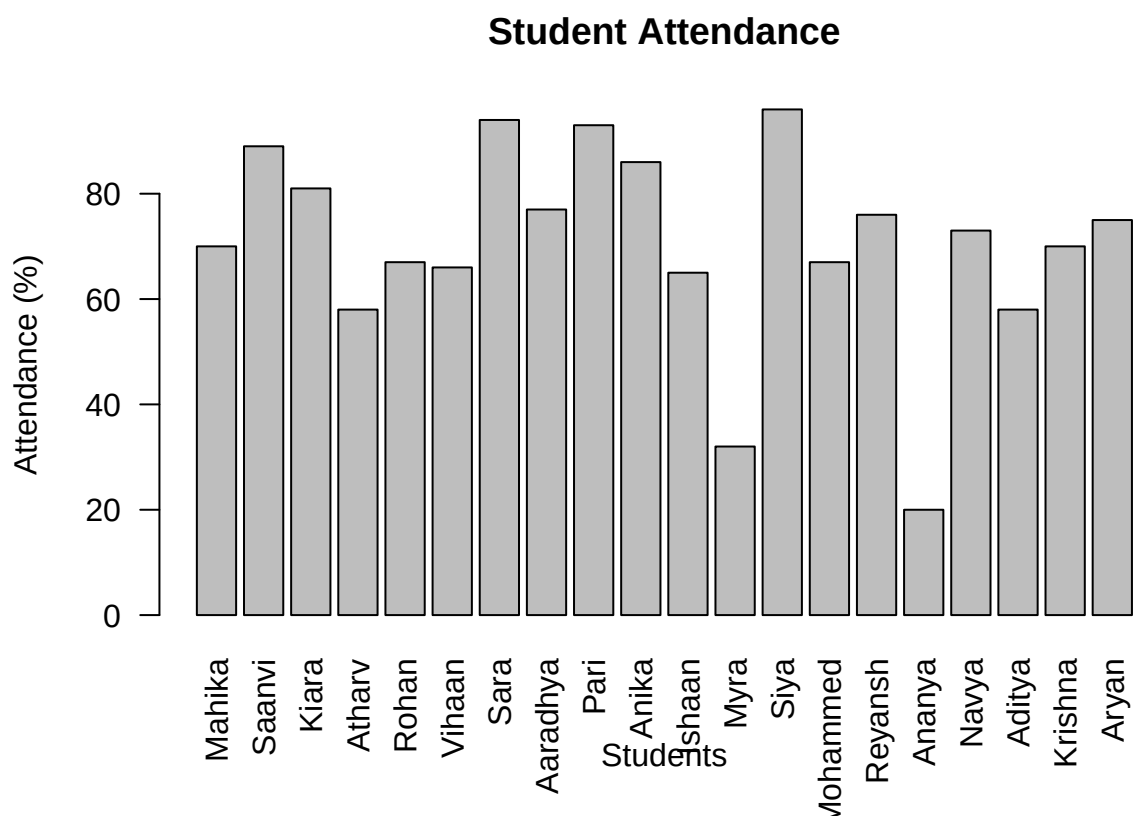
```
barplot(  
  student_data$Marks,  
  names.arg = student_data$Name,  
  main = "Student Marks",  
  xlab = "Students",  
  ylab = "Marks",  
  las = 2  
)
```



```

barplot(
  student_data$Attendance,
  names.arg = student_data$Name,
  main = "Student Attendance",
  xlab = "Students",
  ylab = "Attendance (%)",
  las = 2
)

```



```
cor(
  student_data$Attendance,
  student_data$Marks
)
```

```
## [1] 0.1589152
```

Note that the `echo = FALSE` parameter was added to the code chunk to prevent printing of the R code that generated the plot.